
Topic 15

Configuration Management & Maintenance

1. Version Control

Version control is a system that helps manage changes to software code and documents over time. It allows developers to track modifications, collaborate effectively, and revert to previous versions if necessary.

Types of Version Control Systems (VCS)

- **Local Version Control System (LVCS):** In this system, a local database is used to keep track of file versions. It's simple but not ideal for team collaboration. Example: RCS (Revision Control System).
- **Centralized Version Control System (CVCS):** In this system, there is a central server that contains the files, and each developer has a working copy of the files. Changes are committed to this central repository. Example: SVN (Subversion).
- **Distributed Version Control System (DVCS):** In DVCS, every developer has a local copy of the entire repository. This allows for more flexibility, offline work, and better branching and merging. Example: Git, Mercurial.

Version Control Workflow

- **Commit:** Changes are saved in the repository.
- **Push:** Changes are uploaded to the central repository.
- **Pull:** New changes from the central repository are retrieved to the local working copy.
- **Branching:** Creates a parallel version of the project for independent work.
- **Merging:** Integrates changes from different branches.

Example:

- **Git Workflow Example:**
 1. Clone the repository: `git clone <repository-url>`
 2. Create a new branch: `git checkout -b feature-xyz`
 3. Make changes and commit:
`git add.`
`git commit -m "Added feature xyz"`
 4. Push changes to the remote repository: `git push origin feature-xyz`
 5. Open a pull request to merge changes into the main branch.

2. Maintenance Processes

Software maintenance is the process of updating and improving software after its initial release. Maintenance ensures that the software continues to function as expected, adapts to changes, and fixes issues that arise.

Types of Software Maintenance

- **Corrective Maintenance:** Fixing defects or bugs in the software.
 - Example: A bug causing the app to crash when a user submits a form is identified and fixed.
- **Adaptive Maintenance:** Modifying software to ensure it continues to work in a changing environment.
 - Example: Updating a website to support new versions of a web browser or OS.
- **Perfective Maintenance:** Making improvements or adding new features.
 - Example: Adding a search bar to a website to improve the user experience.
- **Preventive Maintenance:** Making changes to prevent future issues or to improve performance.
 - Example: Refactoring code to improve readability and efficiency, or upgrading the database for faster queries.

Maintenance Activities

- **Bug Fixes:** Addressing and fixing errors reported by users or identified by testers.
- **Updating Dependencies:** Ensuring that all libraries, frameworks, and components used are up-to-date to avoid security vulnerabilities.
- **Optimizing Performance:** Tuning the application to ensure faster response times and efficient resource usage.
- **Documentation Updates:** Maintaining up-to-date documentation to ensure new developers or team members can easily understand the codebase.
- **Refactoring:** Improving the internal structure of the code without changing its external behavior.

Example of a Maintenance Process:

A company develops an e-commerce website. After the initial release:

1. **Corrective Maintenance:** A user reports a bug where the checkout button doesn't work. The development team fixes the bug and releases an update.
 2. **Adaptive Maintenance:** New tax laws require changes to the checkout process. The development team updates the software to reflect the new tax rates.
 3. **Perfective Maintenance:** The development team adds a wish list feature, allowing users to save items for later.
 4. **Preventive Maintenance:** The team identifies code that could be optimized for faster page load times and refactors it.
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Scenario-Based Questions

Scenario 1:

You are managing a project that uses Git for version control. Two developers have made changes to the same file (index.html) on different branches, and now they are trying to merge their branches. However, Git reports a conflict.

Questions:

1. What steps should the developers take to resolve this conflict?
2. How can they prevent similar conflicts in the future?

Solution:

1. **Steps to resolve the conflict:**
 - The developers need to manually review the conflict in the file (index.html) and decide which changes to keep.
 - After resolving the conflict, they should mark the file as resolved by staging it: `git add index.html`.
 - Finally, commit the merge: `git commit -m "Resolved merge conflict in index.html"`.
 2. **Preventing future conflicts:**
 - Communicate more frequently about which parts of the code each developer is working on.
 - Use smaller, more frequent commits to avoid large, conflicting changes.
 - Implement a clear branching strategy (e.g., feature branches, development branches).
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Scenario 2:

Your team is maintaining a web application. The application's performance has degraded after a new feature was added, and users are complaining about slow load times.

Questions:

1. What steps would you take to identify and fix the performance issue?
2. What kind of maintenance is this an example of?

Solution:

1. **Steps to identify and fix the performance issue:**
 - Profile the application to identify slow functions or resource bottlenecks.
 - Use browser developer tools to inspect network activity and loading times.
 - Optimize queries, reduce large image sizes, and improve caching strategies.
 - Test the application's performance under various conditions (e.g., high traffic).
 2. **Type of maintenance:**

This is an example of **Preventive Maintenance**, as you are improving the performance to prevent future issues and enhance user experience.
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Scenario 3:

You are managing a project using SVN for version control. A new developer has joined your team and is unsure how to make their first commit.

Questions:

1. What steps should the new developer follow to make their first commit?
2. What should the developer do if they encounter a merge conflict?

Solution:

1. Steps for making the first commit:

- Check out the project from the SVN repository: `svn checkout <repository-url>`.
- Make changes to the files locally.
- Stage the changes: `svn add <file-path>` (if the file is new).
- Commit the changes: `svn commit -m "Commit message"`.

2. Handling a merge conflict:

- The developer needs to manually resolve the conflict by reviewing the conflicting files.
 - Once resolved, the developer should mark the conflict as resolved: `svn resolved <file-path>`.
 - Commit the changes again to the repository: `svn commit -m "Resolved merge conflict"`.
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